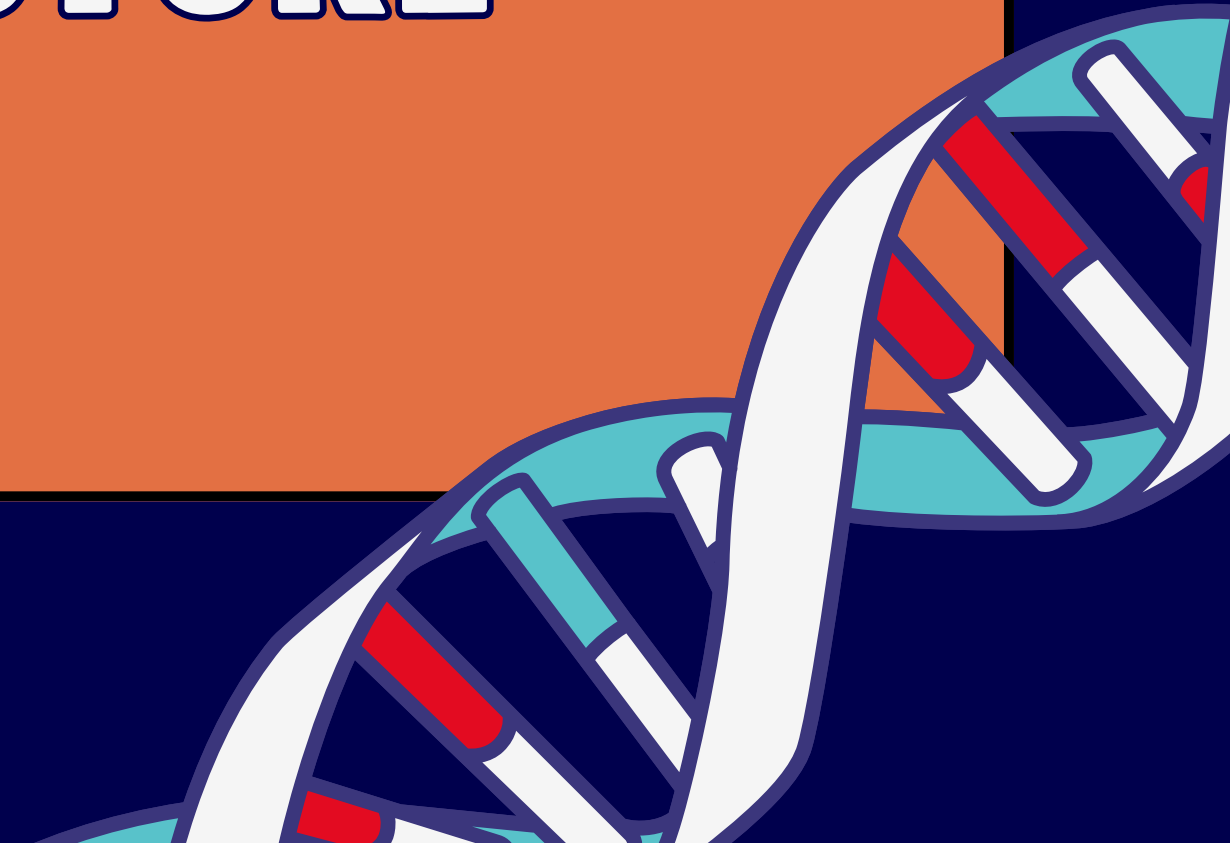




KARYOTYPING DIFFERENT ABNORMAL CASES III

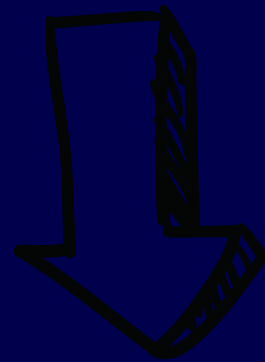
UNBALANCED STRUCTURE

Adan Alnaamneh



Chromosome Abnormalities

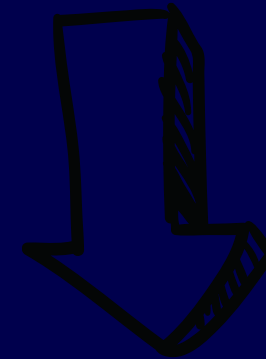
Numerical



Polyploidy

Aneuploidy

Structural

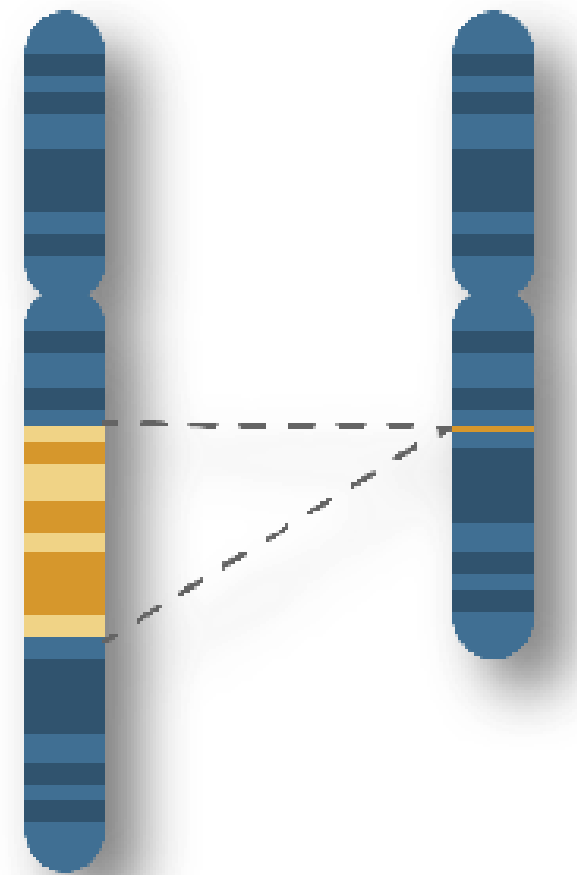


Balanced

Unbalanced

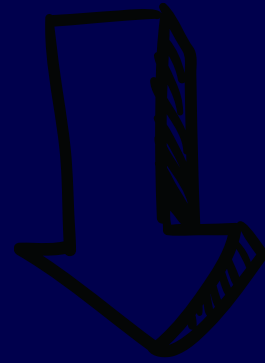
DELETION

- Deletion: is a genetic aberration in which a part of a chromosome or a sequence of DNA is left out during DNA replication
- Loss of a (generally small) segment of chromosome



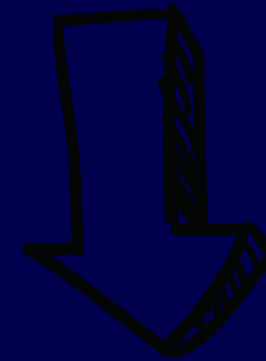
Deletion

Terminal

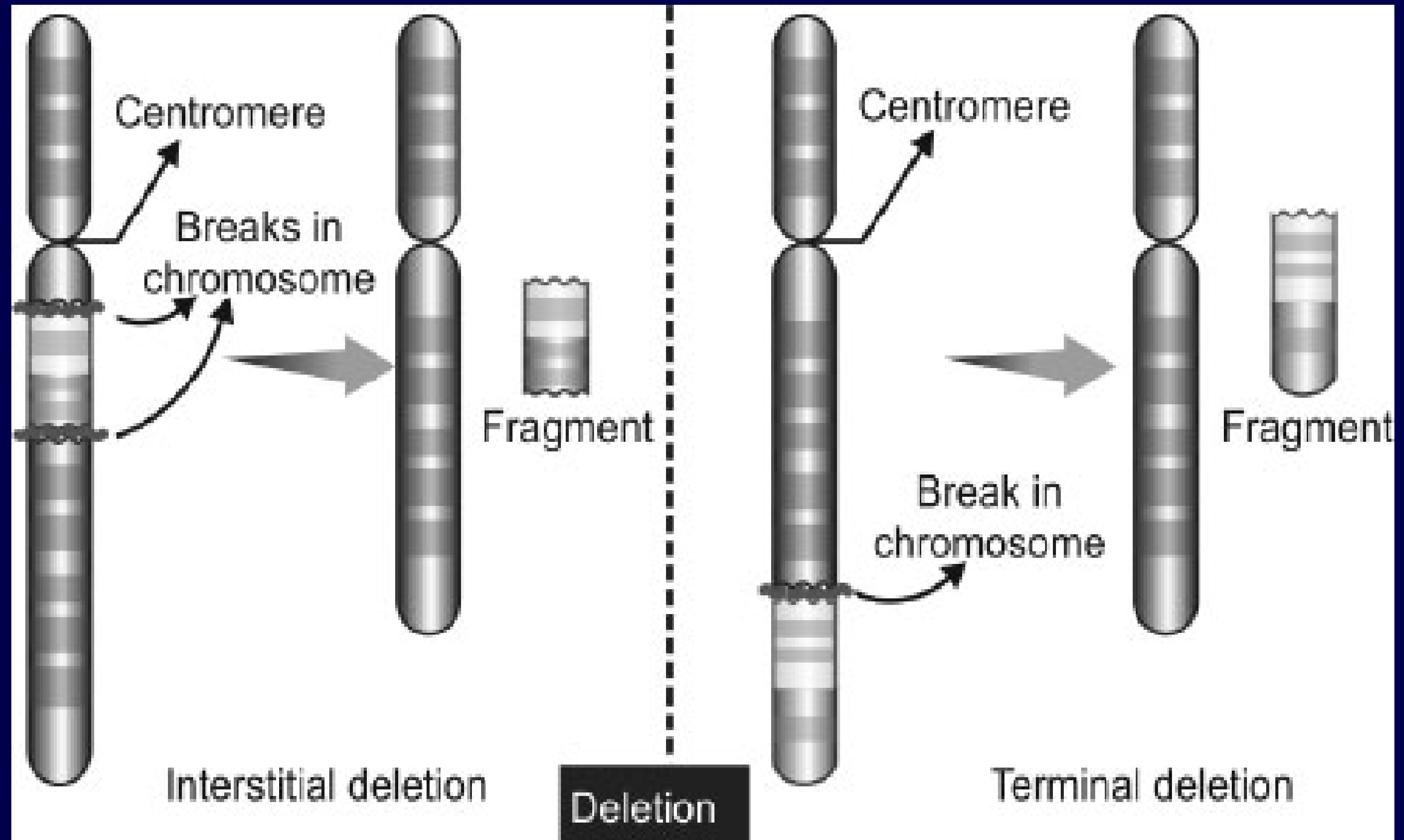


that occur at the end of a chromosome.

interstitial



that occur along a chromosome arm.



Deletion in Human

Cri-du-chat syndrome

Deletion in chromosome 5

Angelman syndrome

Deletion of chromosome 15

William syndrome

Deletion in chromosome 7

Prader-Willi syndrome

Deletion of chromosome 15

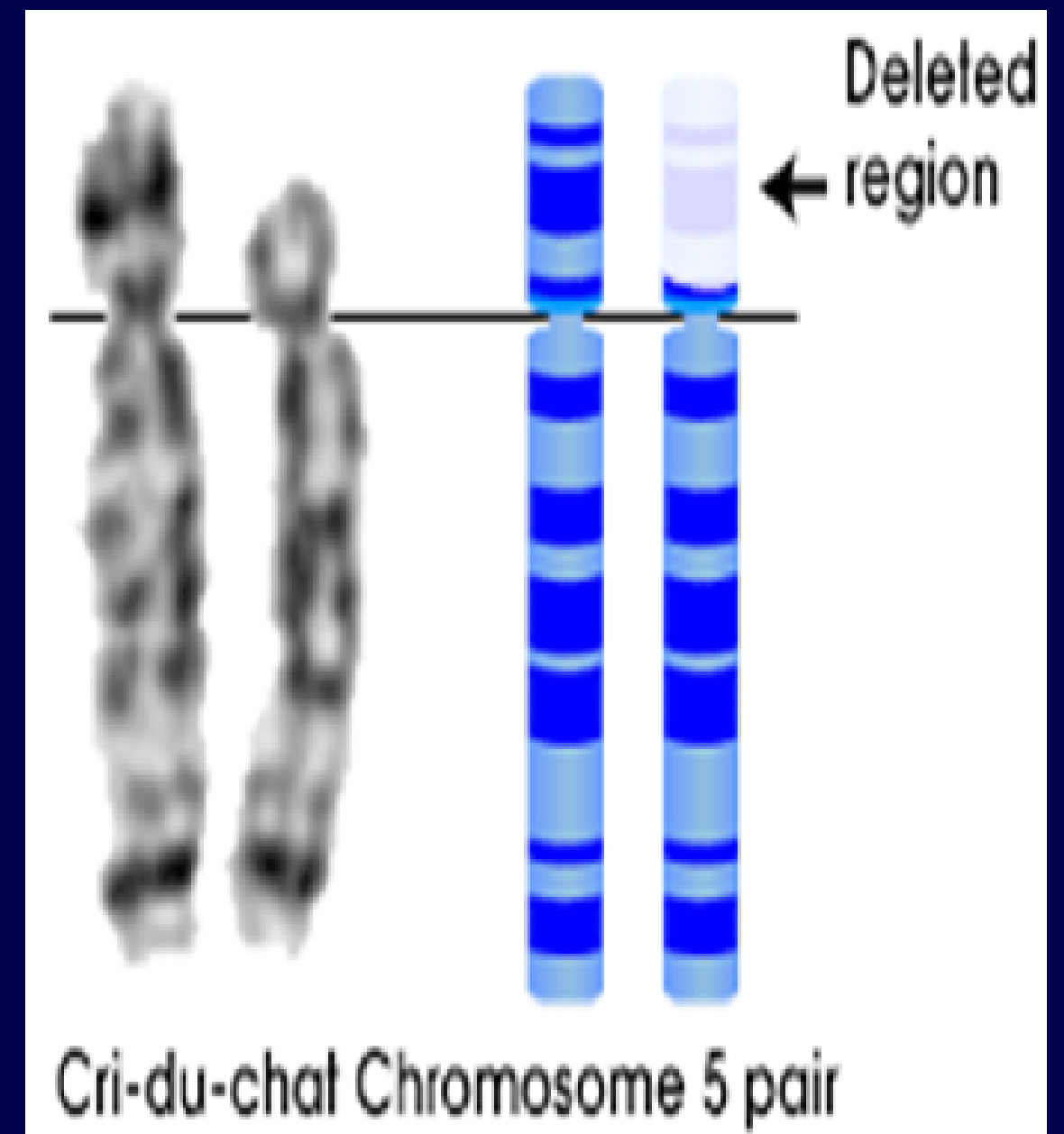
Deletion

Terminal deletion example:

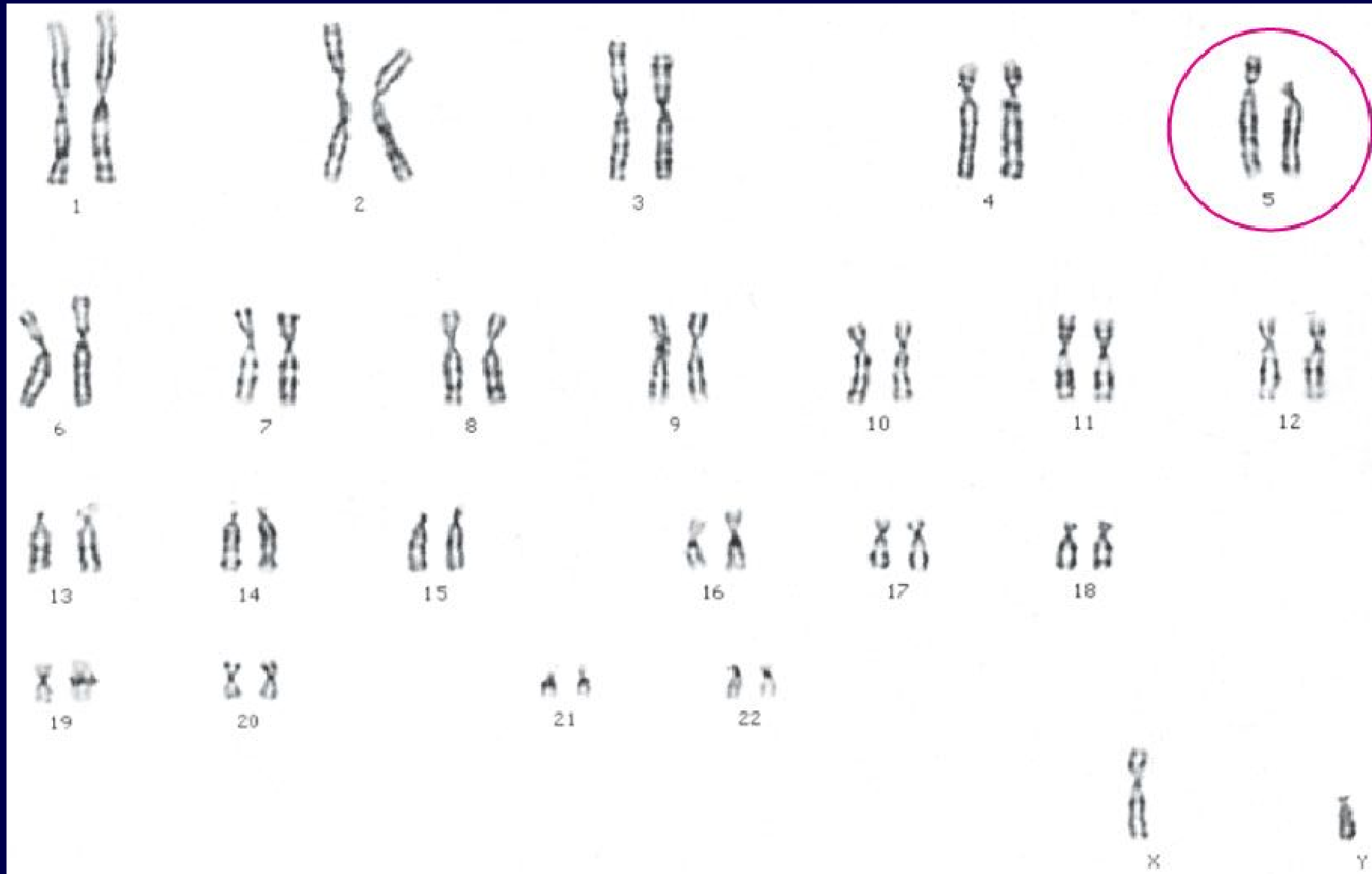
Cri-du-chat syndrome:

Karyotype:

- 46,XX,del(5) (p15.2) or
- 46,XX,-5p



Cri-du-chat syndrome



Cri-du-chat syndrome

- 1st autosomal deletion described
- Symptoms:
 1. cat-like cry, which disappears with age
 2. Severe mental retardation
 3. Congenital heart disease
 4. Hypertelorism (widely separated eyes)
 5. Low birth weight and poor growth
 6. Severe cognitive, speech, and motor delay
 7. Behavioral problems



Deletion

Interstitial deletion example:

William syndrome:

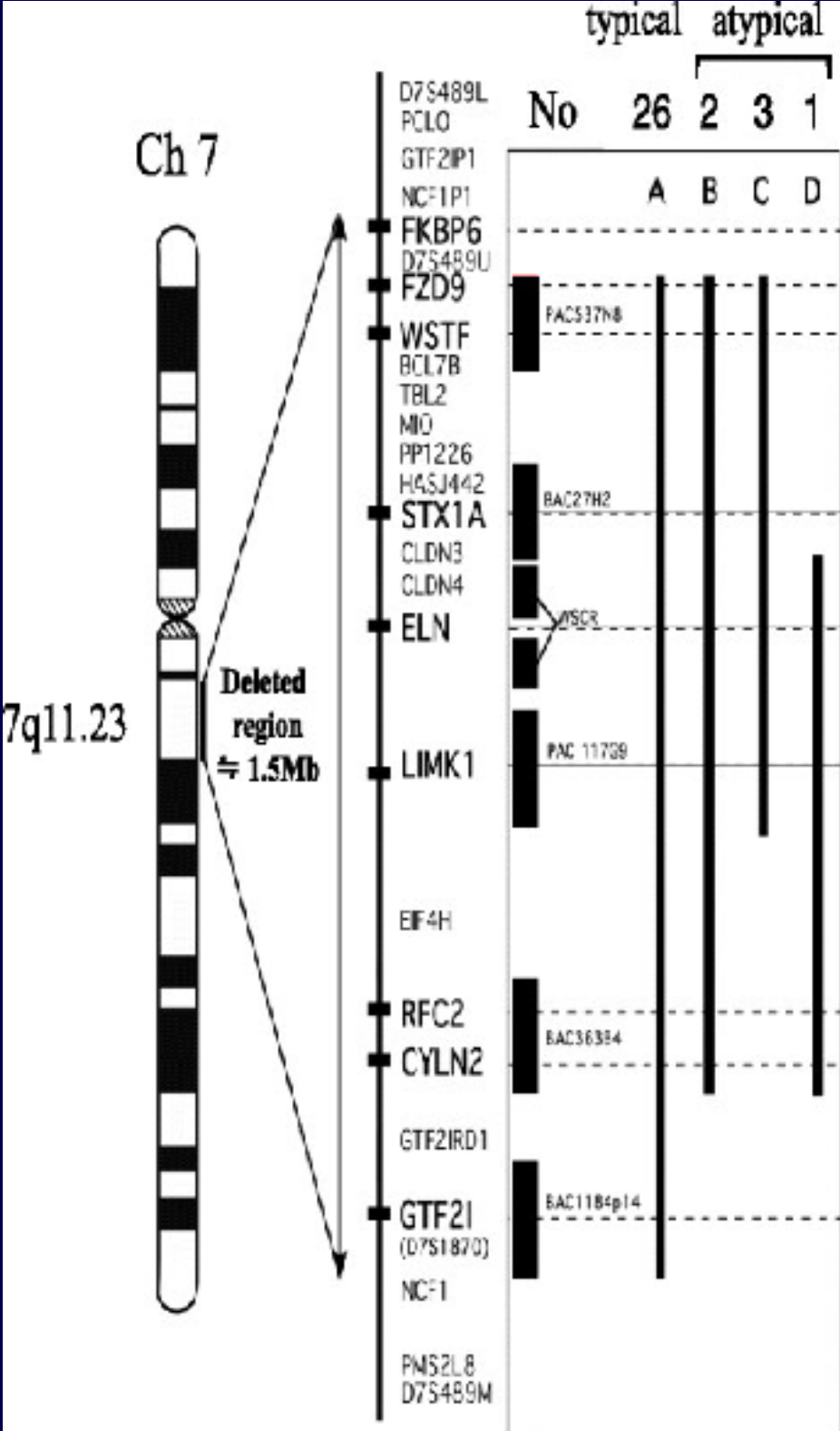
46,XX,del(7) (q11.23)

- deletion happen only in the band

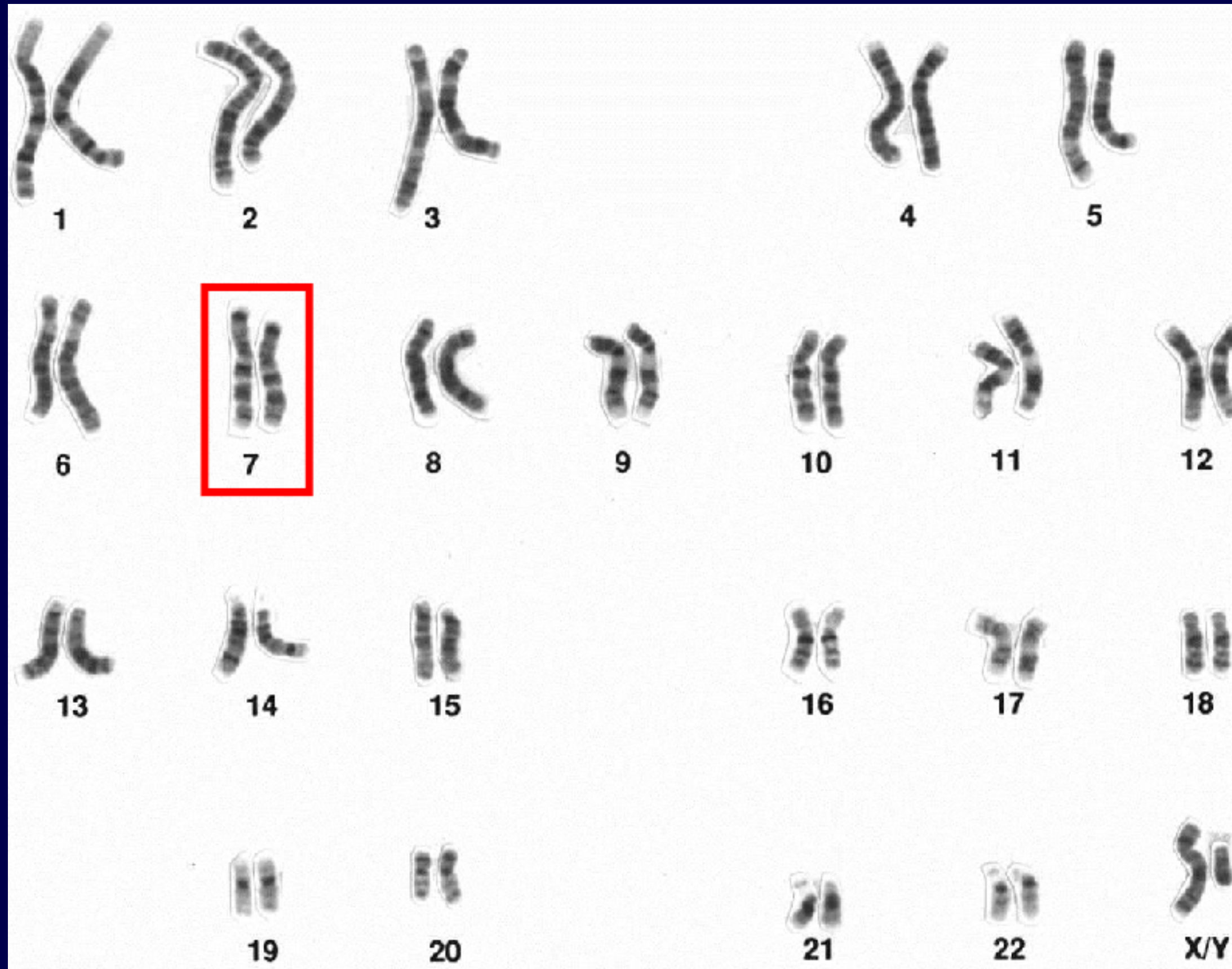
William syndrome:

46, XX, del(7) (q11.23)

- deletion happen only in the band



William syndrome



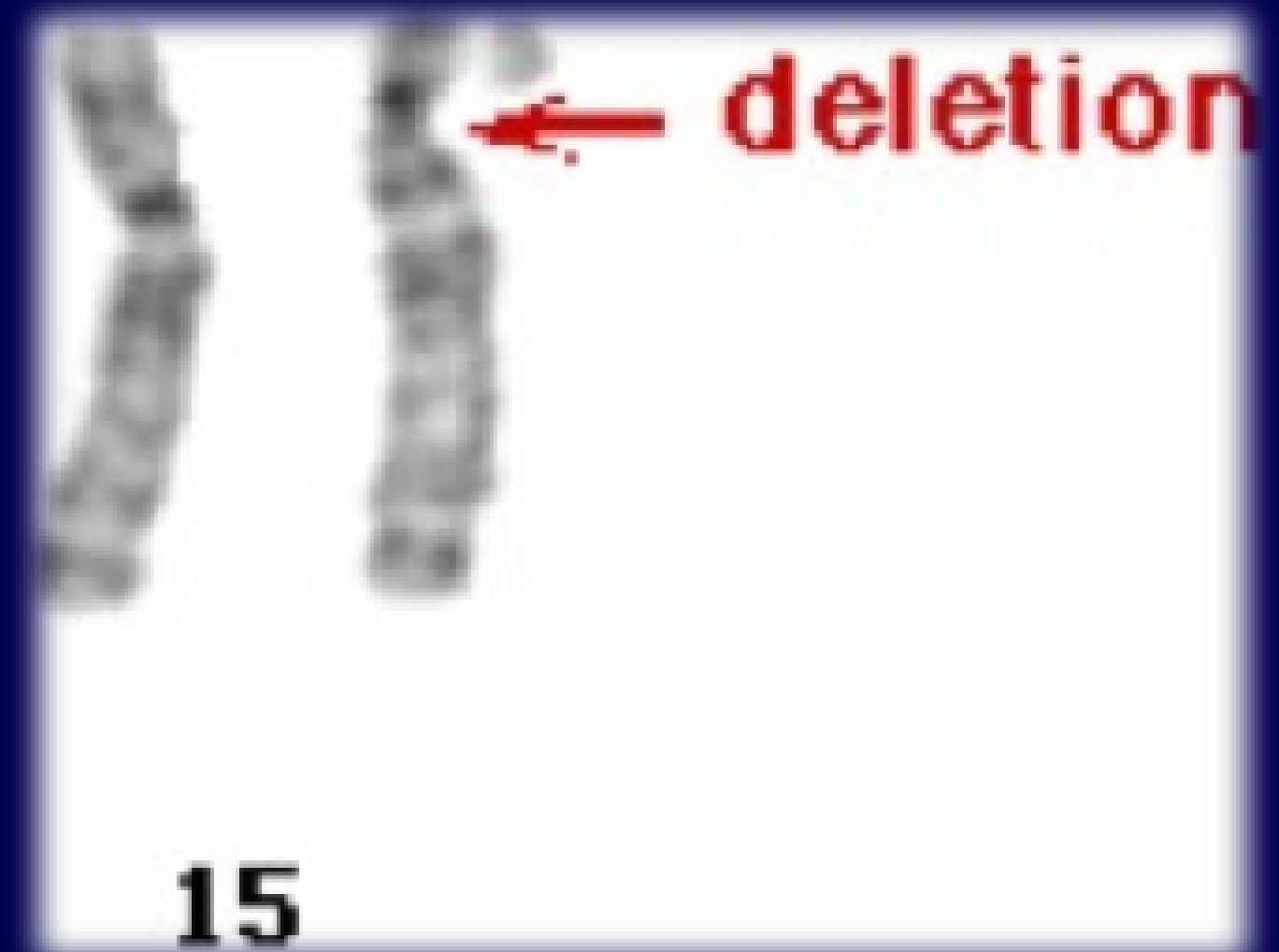
William syndrome

- Symptoms:
- Feeding problems, including colic, reflux, and vomiting
- Heart disease or blood vessel problems
- Developmental delay, mild to moderate intellectual disability, learning disorders, and Delayed speech
- Easily distracted
- A flattened nasal bridge with small upturned nose
- Prominent lips with an open mouth
- 1. Partially missing teeth, or small, widely spaced teeth

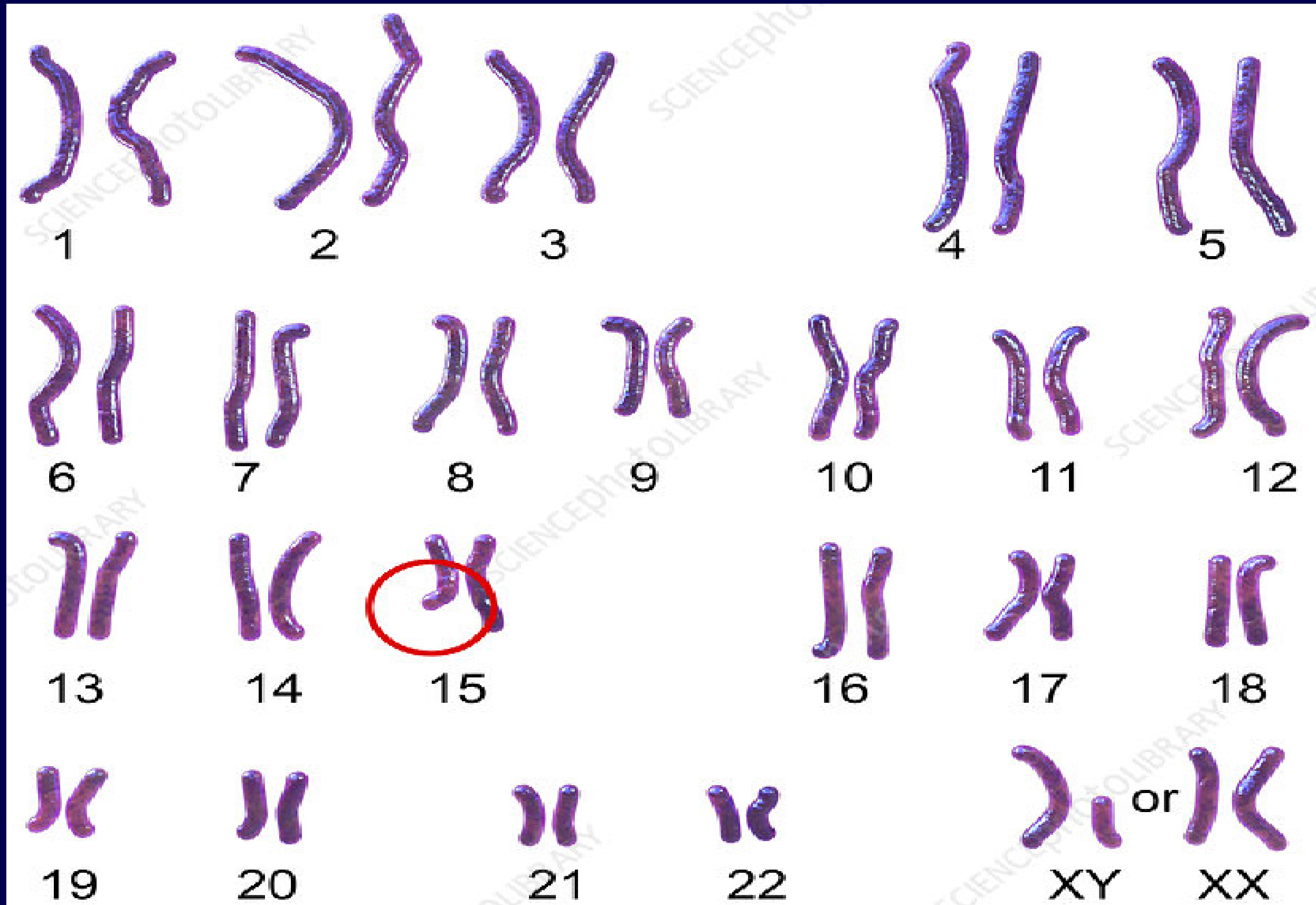


Deletion

- Angelman syndrome
46,XX,del(15) (q11) Mat
- Prader-Willi syndrome
46,XX,del(15) (q11) Pat



Angelman syndrome or Prader-Willi



Angelman syndrome

- Symptoms:

1. Developmental delays & Intellectual disability
2. No speech or minimal speech
3. Difficulty walking, moving or balancing well
4. Frequent smiling and laughter, excitable personality
5. Sucking or feeding difficulty
6. Small head size, with flatness in the back of the head
7. Unusual behaviors, like hand flapping and arms uplifted while walking
8. Curved spine (scoliosis)



Prader-Willi Syndromes

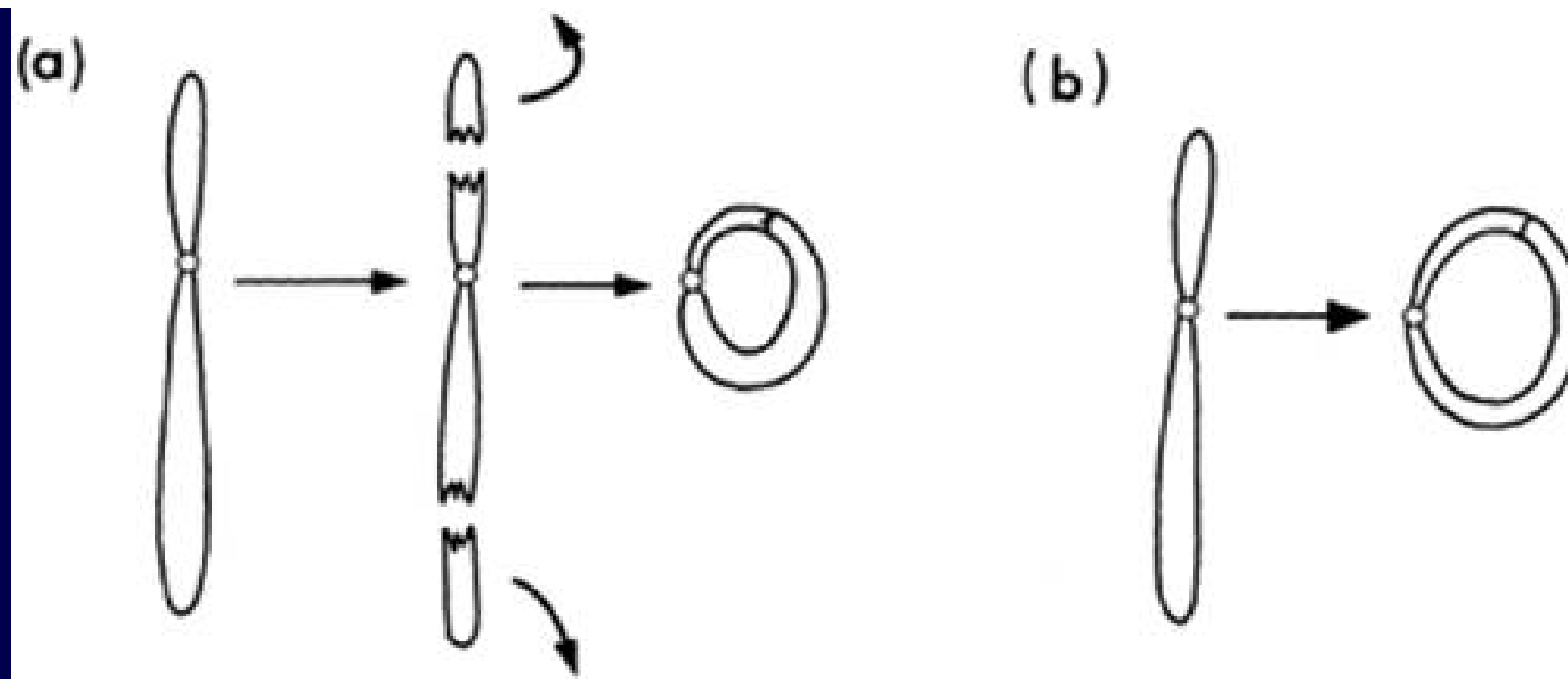
- Symptoms:
- Poor muscle tone, responsiveness, growth and physical development
- Distinct facial features, almond-shaped eyes, a turned-down mouth and a thin upper lip.
- Underdeveloped genitals.
- Food craving and weight gain.
- Speech problems & Behavioral problems.
- small hands and feet, hip problems, reduced saliva flow

Hypotonia
(decreased
muscle tone)



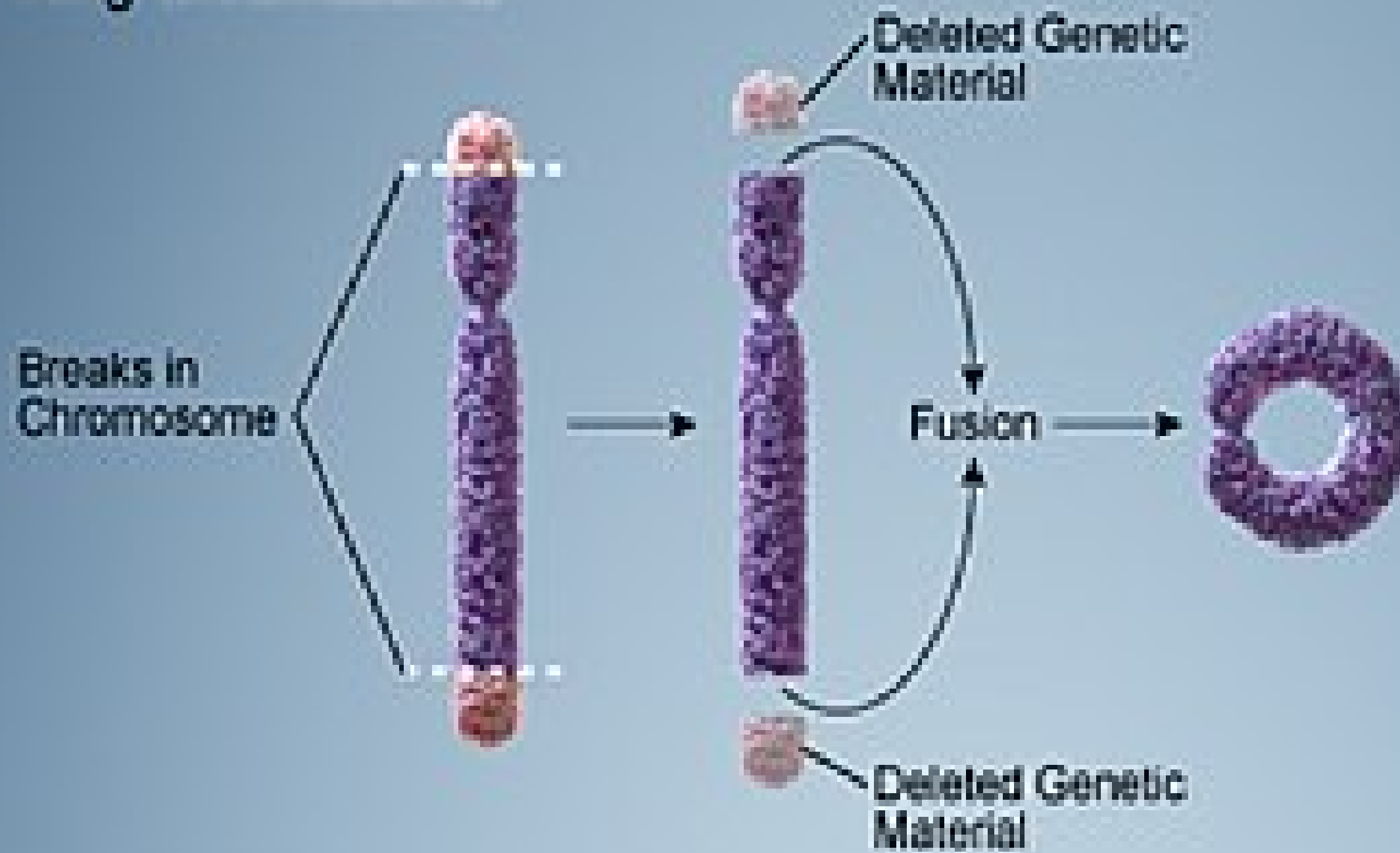
Ring Chromosome

- Ring chromosome: is a circular structure that occurs when a chromosome breaks in two places and its broken ends fuse together.
- Occurs due to loss of both ends of a chromosome
- The broken ends rejoin to form a ring-like chromosome



Ring Chromosome

Ring Chromosome



Ring Chromosome

- 46,XX,r(15) (p11q26)
- Symptoms:
 1. physical and mental development
 2. poor speech development
 3. recurrent pneumonia
 4. Heart problems
 5. Expressionless face

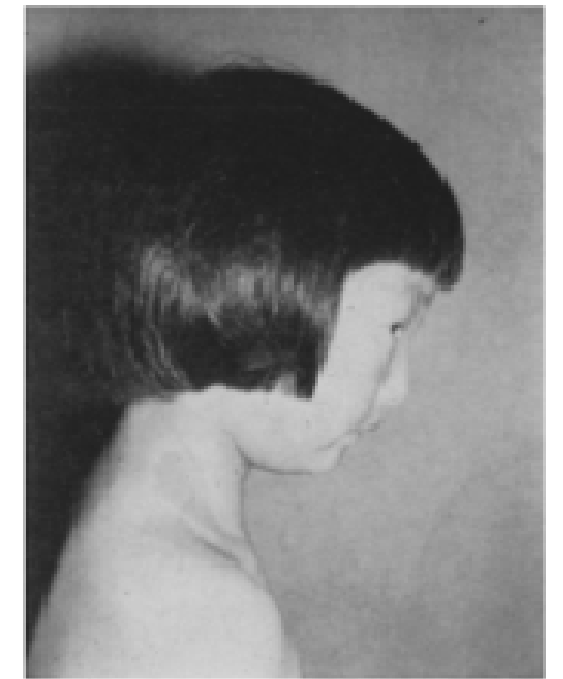
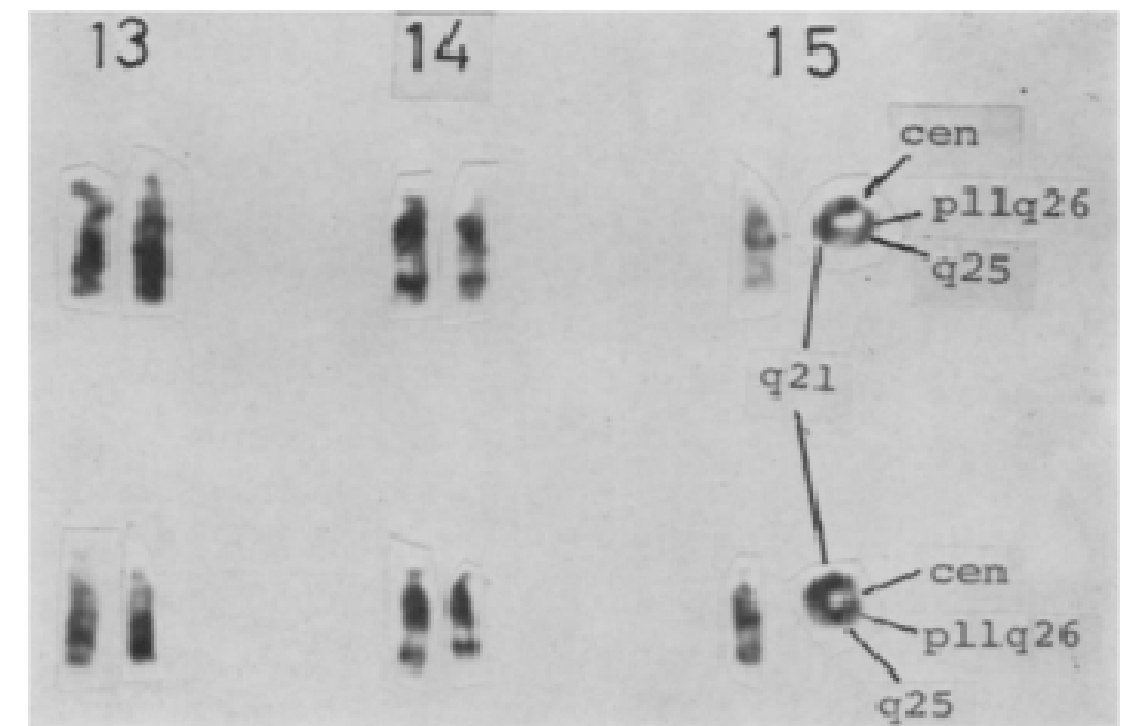
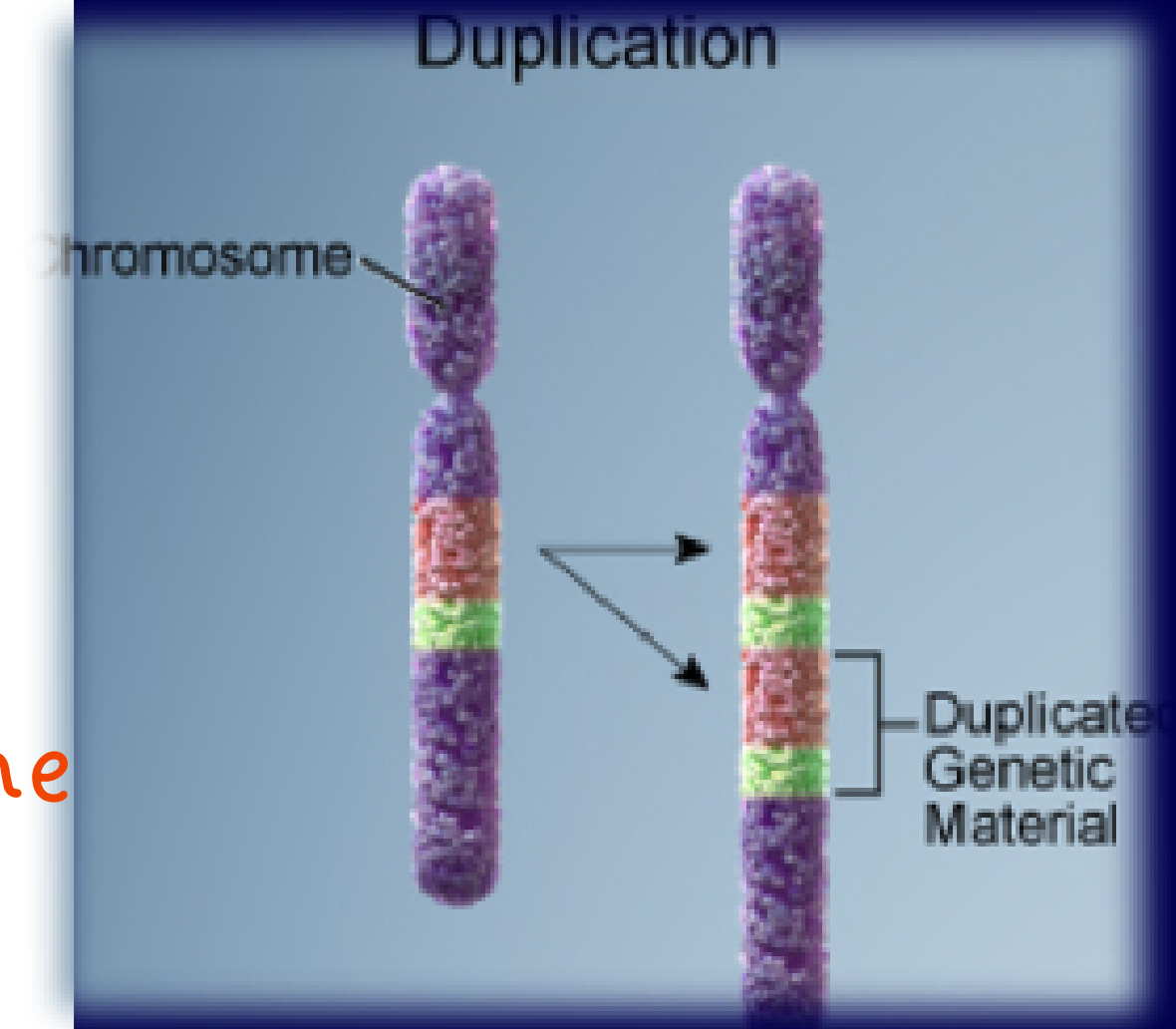


Fig. 1. The patient at five and a half years of age.



Duplication

- Duplications: Rearrangements that result in an extra copy of a chromosome.
- The duplicated material can be:
 1. Located in tandem (side-by-side)
 2. Positioned at a separate site either within the same chromosome or within a different chromosome.



Duplication

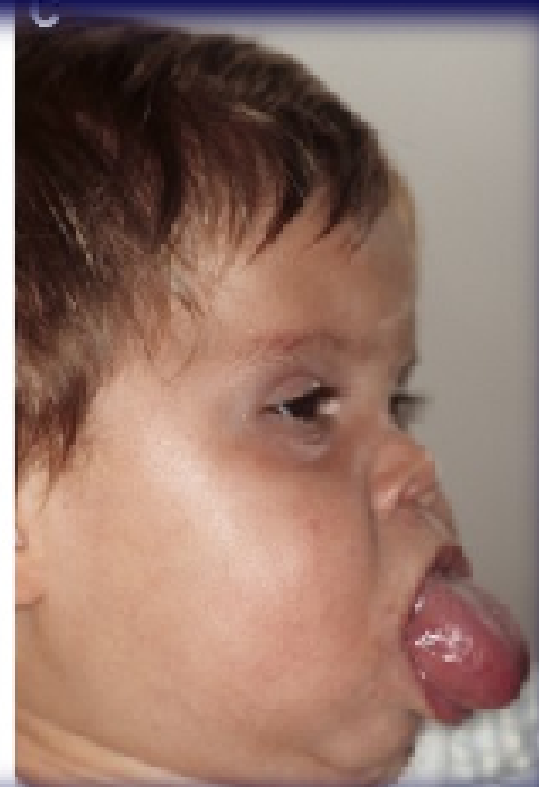
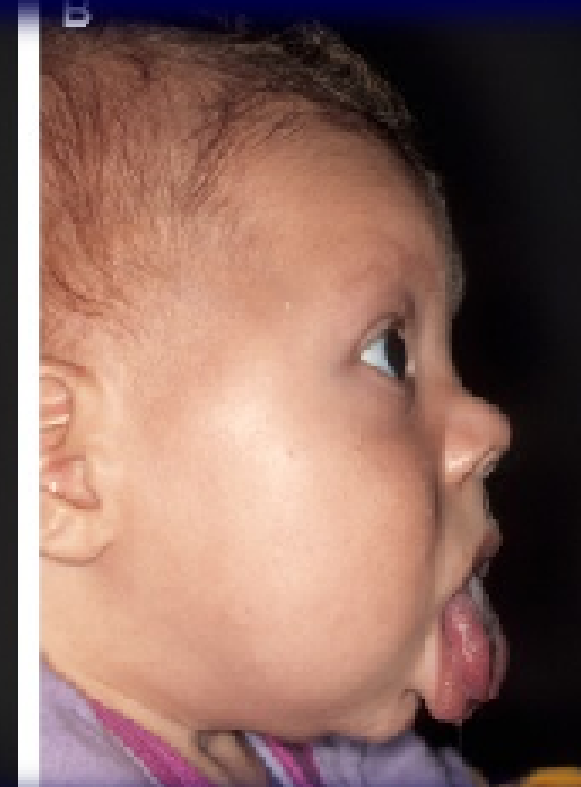
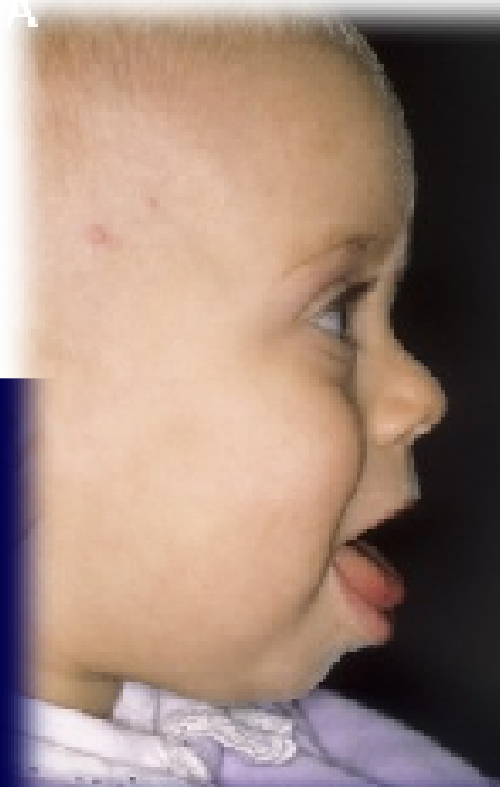
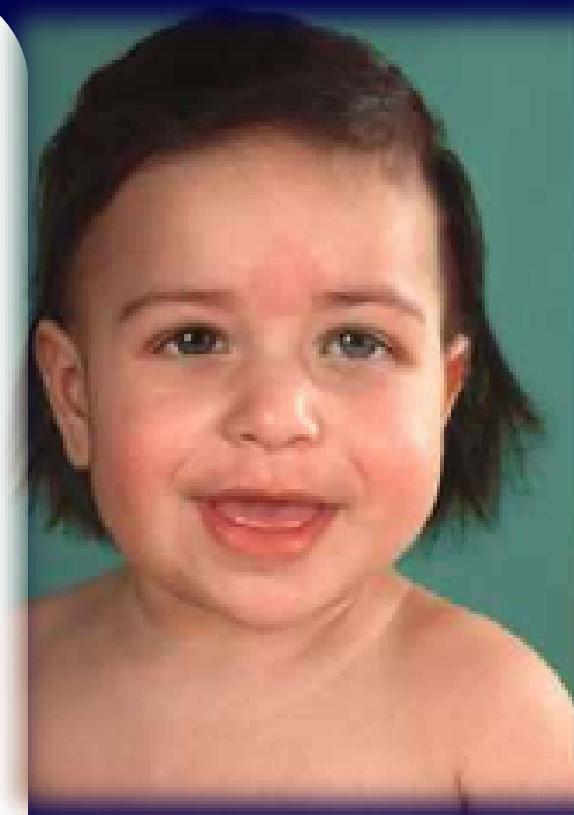
- Beckwith-wiedemann syndrome
46,XX,dup(11)(11p15. 5)



Beckwith-wiedemann syndrome ?

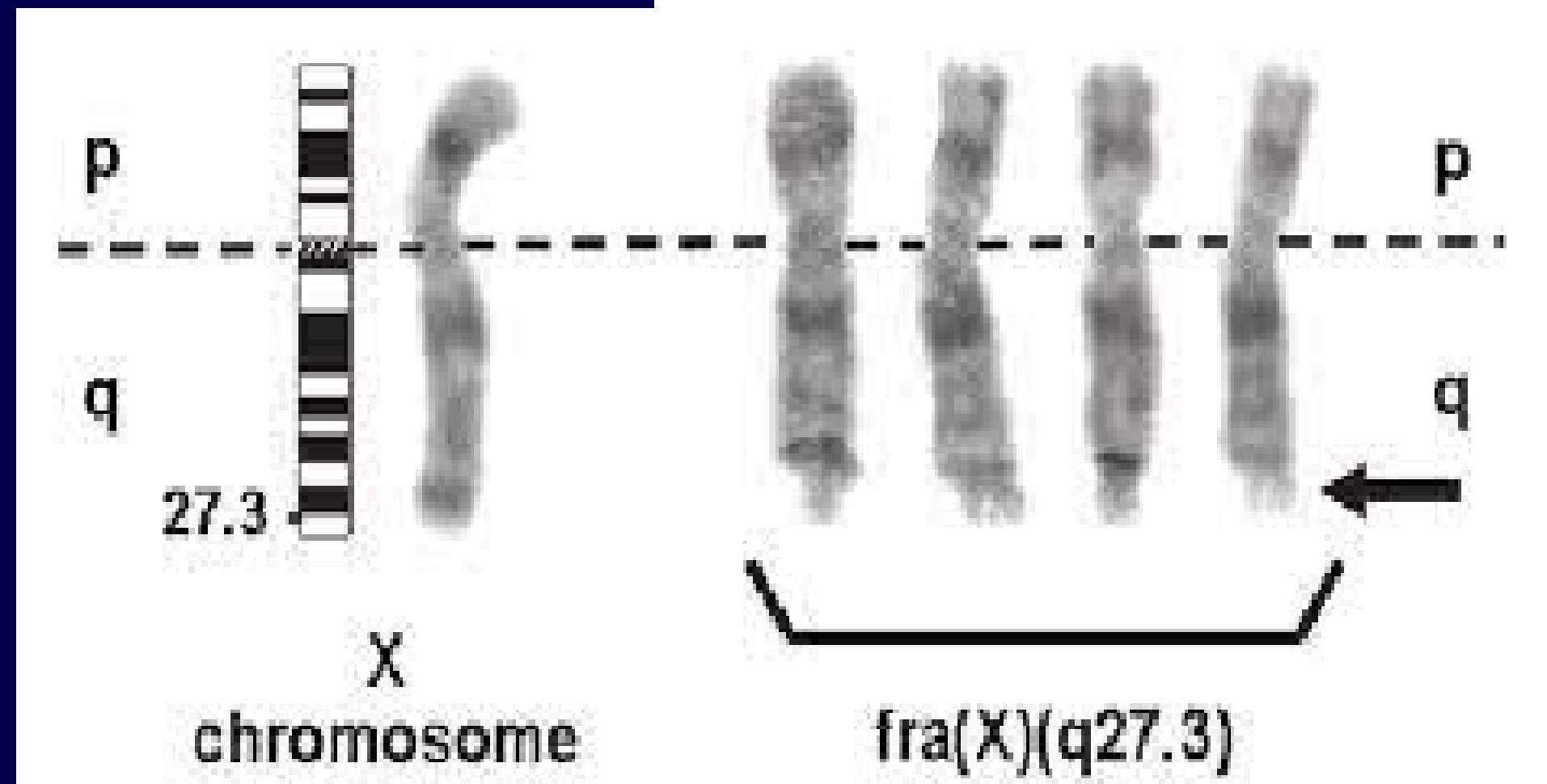
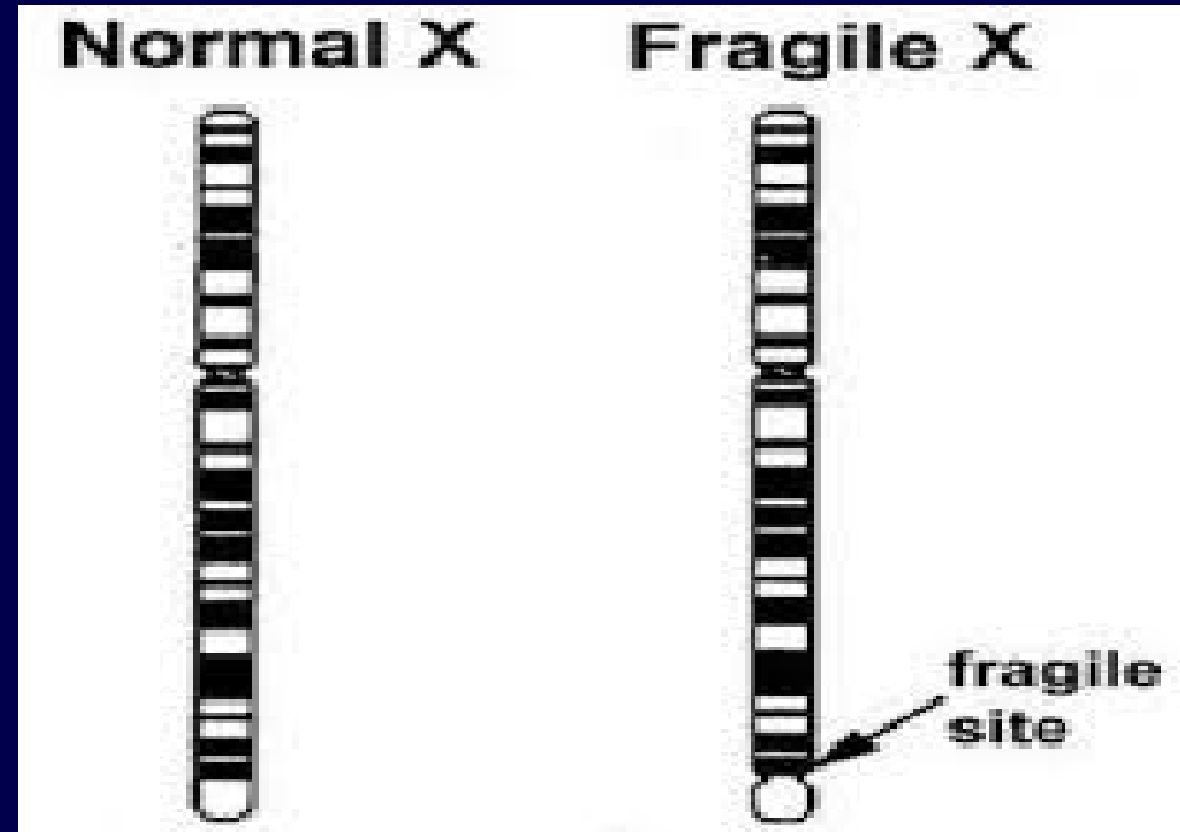
- Symptoms:

1. Overgrowth of one side of the body or a part of one side of the body
2. An enlarged tongue
3. Low blood sugar & hypoglycemia
4. Enlarged organs, such as the kidneys, liver, or pancreas
5. Creases in the earlobe or behind the ear



Duplication

- Fragile X syndrome
 $46,X,fra(x)(q27.3)$



Fragile X syndrome

- Symptoms:

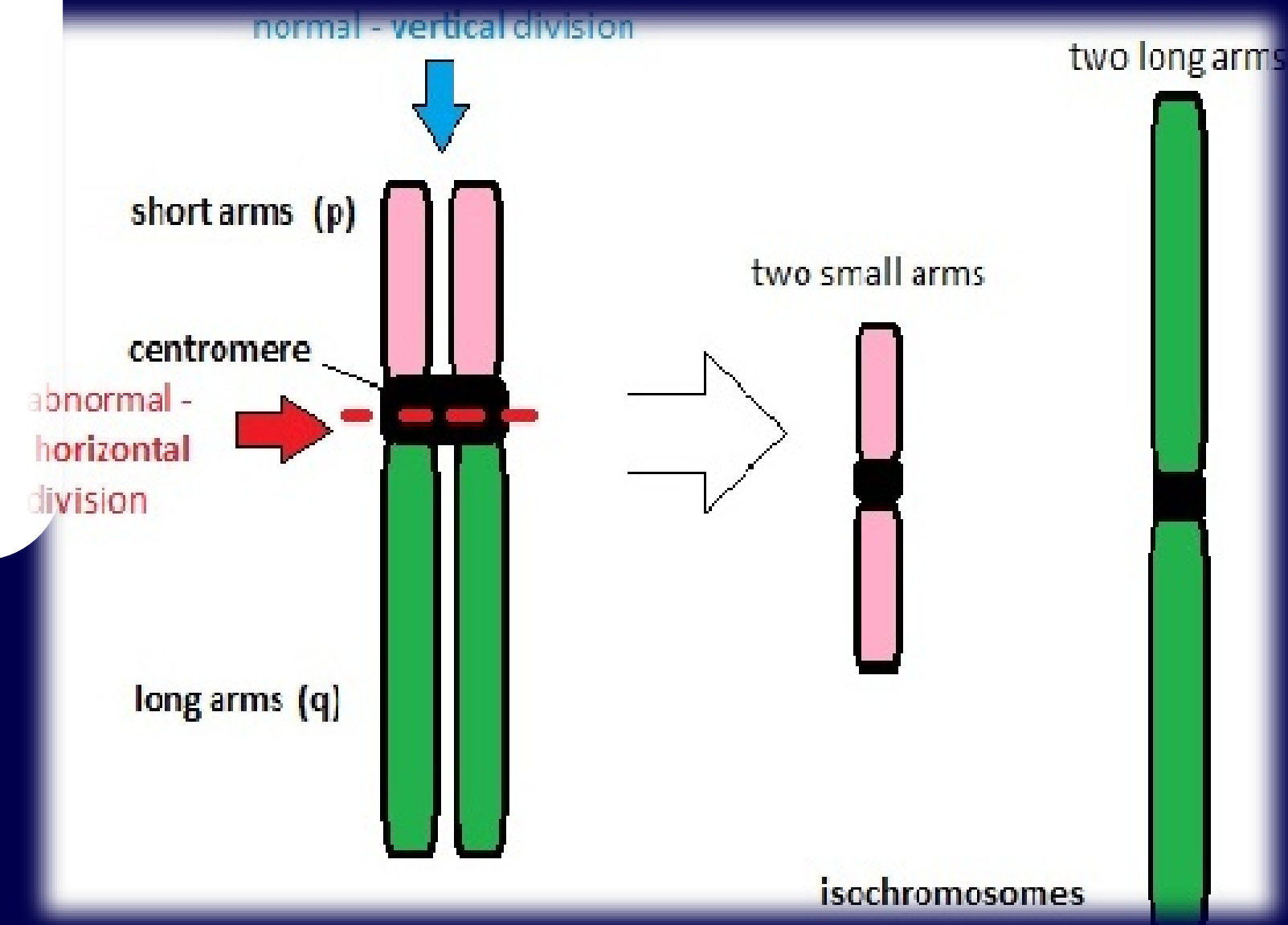
1. Long and narrow face
2. Large ears, a prominent jaw and forehead
3. Unusually flexible fingers and flat feet
4. Enlarged testicles in males after puberty.



Isochromosome

- Isochromosome:

An abnormal chromosome in which one arm is duplicated (forming two arms of equal lengths, with the same loci in reverse sequence) and the other arm is missing.



Turner syndrome

- Symptoms:
- Swollen hands and feet
- Thick neck tissue and swelling of the neck
- A broad chest and widely spaced nipples
- A large number of moles
- Lazy eyes & droopy eyelids
- Heart conditions & kidney abnormalities
- Delayed sexual development
- Underactive thyroid gland & high blood pressure & diabetes
- Teeth problems



- 47,XY,+21,t(1;4) (q11;q54)
- 46,XX,t(X;5) (p31;p51)